

STT 465 – Homework 2: Single-Parameter Bayesian Models

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Question 1: Bernoulli-Beta Posterior

Let $s = \sum x_i = 18$, $n = 25$, so $n - s = 7$.

(a) $L(\omega | x) = \prod_{i=1}^{25} \omega^{x_i} (1 - \omega)^{1-x_i} \propto \omega^{18} (1 - \omega)^7$.

(b) The prior is $\pi(\omega) \propto \omega(1 - \omega)$. By Bayes' theorem

$$\pi(\omega | x) \propto L(\omega | x) \pi(\omega) \propto \omega^{18+1} (1 - \omega)^{7+1} = \omega^{20-1} (1 - \omega)^{9-1},$$

the kernel of a Beta distribution, so $\omega | x \sim \text{Beta}(20, 9)$.

(c) For $\text{Beta}(\alpha, \beta)$, $E = \alpha / (\alpha + \beta)$. In general the posterior is $\text{Beta}(a + s, b + n - s)$, so

$$E[\omega | x] = \frac{a + s}{a + b + n} = \frac{2 + 18}{2 + 2 + 25} = \frac{20}{29} \approx 0.6897.$$

(d) Mode = $\frac{\alpha - 1}{\alpha + \beta - 2} = \frac{19}{27} \approx 0.7037$. Median: `qbeta(0.5, 20, 9)  $\approx$  0.6941.`

(e)

```
1 - pbeta(0.70, 20, 9) # equivalently pbeta(0.70, 20, 9, lower.tail = FALSE)
```

$$P(\omega > 0.70 | x) \approx 0.4725.$$

Question 2: Prior Sensitivity and Bayesian Updating

Here $n = 21$, $x = 18$ (successes), $n - x = 3$.

(a) Posterior $\propto \omega^{18} (1 - \omega)^3 \cdot \omega^0 (1 - \omega)^0$, so $\omega | x \sim \text{Beta}(1 + 18, 1 + 3) = \text{Beta}(19, 4)$.

(b) Posterior $\propto \omega^{18} (1 - \omega)^3 \cdot \omega^3 (1 - \omega)^{-0.1}$, so $\omega | x \sim \text{Beta}(4 + 18, 0.9 + 3) = \text{Beta}(22, 3.9)$.

(c) The posterior is $\text{Beta}(a + x, b + n - x)$, so

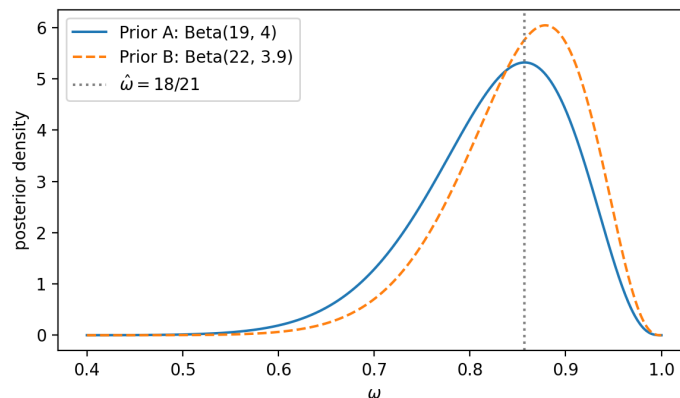
$$E[\omega | x] = \frac{a + x}{a + b + n} = \frac{a + b}{a + b + n} \cdot \frac{a}{a + b} + \frac{n}{a + b + n} \cdot \frac{x}{n}.$$

The first term is the prior mean $a/(a + b)$ weighted by $(a + b)/(a + b + n)$ (the prior “sample size” share); the second is the sample proportion x/n weighted by $n/(a + b + n)$ (the data’s share). The posterior mean is a weighted average of prior mean and MLE.

- (d) Prior A: $19/23 \approx 0.8261$. Prior B: $22/25.9 \approx 0.8494$. They differ because Prior B has prior mean $4/4.9 \approx 0.816$ and prior weight $a + b = 4.9$ versus 2 for Prior A (prior mean 0.5). Both are pulled from the sample proportion $18/21 \approx 0.857$ toward their prior mean, but A is pulled more, since it has a lower prior mean and a less informative prior, while B sits closer to the data. With $n = 21$ the data dominate, so the difference is small.

(e)

```
w <- seq(0, 1, length.out = 1000)
plot(w, dbeta(w, 19, 4), type = "l", lwd = 2, col = "blue",
      xlab = expression(omega), ylab = "Posterior density")
lines(w, dbeta(w, 22, 3.9), lwd = 2, col = "red", lty = 2)
abline(v = 18/21, lty = 3)
legend("topleft", c("Prior A: Beta(19,4)", "Prior B: Beta(22,3.9)", "18/21"),
      col = c("blue","red","black"), lty = c(1,2,3), lwd = 2)
```



Both posteriors are concentrated around 0.8–0.9 and overlap heavily. Posterior B is shifted slightly right (closer to $18/21$), while Prior A's posterior has slightly more mass at lower values and is slightly wider. With 21 observations the choice of prior matters little.

Question 3: Credible Intervals and HPDI

(a)

```
qbeta(c(0.025, 0.975), shape1 = 22, shape2 = 4)
```

The 95% equal-tailed interval is $[0.6878, 0.9546]$.

- (b) Given the model, prior and observed data, there is a 95% posterior probability that ω lies in $[0.688, 0.955]$. Unlike a frequentist confidence interval, ω is treated as random and the interval as fixed once the data are observed.

(c)

```
library(HDIInterval)
set.seed(1)
omega_post <- rbeta(10000, shape1 = 22, shape2 = 4)
hdi(omega_post, credMass = 0.95)
```

The 95% HPDI is approximately $[0.705, 0.964]$ (exact HPD from the density: $[0.709, 0.967]$).

- (d) The equal-tailed interval has length 0.2668; the HPDI has length about 0.26 (exact 0.2573). The HPDI is shorter.
- (e) The equal-tailed interval leaves 2.5% in each tail, while the HPDI is the shortest interval with 95% mass, containing the highest-density points. For a symmetric unimodal posterior they coincide. Beta(22, 4) is left-skewed, with a long lower tail and a steep upper end, so the HPDI shifts right and trims the low-density lower tail.

Question 4: Comparing Two Population Proportions

- (a) With uniform priors, $\omega_x \mid x \sim \text{Beta}(1 + 437, 1 + 543) = \text{Beta}(438, 544)$ and $\omega_y \mid y \sim \text{Beta}(1 + 398, 1 + 422) = \text{Beta}(399, 424)$, independently.
- (b) $E[\omega_x \mid x] = 438/982 \approx 0.4460$; $E[\omega_y \mid y] = 399/823 \approx 0.4848$.

(c)

```
set.seed(1)
U <- 10000
omega_x <- rbeta(U, shape1 = 438, shape2 = 544)
omega_y <- rbeta(U, shape1 = 399, shape2 = 424)
mean(omega_x < omega_y)
```

Estimate: ≈ 0.950 (an accurate large-simulation value is 0.950).

- (d) It is the posterior probability, given both samples, that the female birth proportion in group 1 is lower than in group 2: our degree of belief in that ordering after seeing the data.
- (e) $P(\omega_x < \omega_y \mid x, y) \approx 0.95$ is a direct probability statement about the parameters, conditional on the observed data and priors. The `prop.test` p-value (about 0.052 with continuity correction) is the probability, *assuming* $\omega_x = \omega_y$, of seeing a difference at least this extreme in repeated sampling. It is not the probability that the null (or alternative) is true, so the two numbers answer different questions even though they are numerically close here.

Question 5: Poisson-Gamma Model and Prediction

(a) $L(\lambda \mid x) = \prod_i \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{\lambda^{\sum x_i} e^{-n\lambda}}{\prod x_i!} \propto \lambda^{\sum x_i} \exp(-n\lambda).$

(b) $\pi(\lambda) \propto \lambda^{a-1} e^{-b\lambda}$, so

$$\pi(\lambda \mid x) \propto \lambda^{a+\sum x_i-1} e^{-(b+n)\lambda} \Rightarrow \lambda \mid x \sim \text{Ga}\left(a + \sum x_i, b + n\right) = \text{Ga}(56, 21).$$

(c) $E[\lambda \mid x] = 56/21 \approx 2.667.$

```
qgamma(c(0.025, 0.975), shape = 56, rate = 21)
```

95% equal-tailed interval: [2.014, 3.409].

- (d) By the law of iterated expectations, $E[X_f | x] = E[E[X_f | \lambda, x] | x] = E[\lambda | x] = 56/21 \approx 2.667$, since $X_f | \lambda \sim \text{Poi}(\lambda)$ has mean λ and is independent of x given λ .

(e)

```
set.seed(1)
U <- 10000
lambda_post <- rgamma(U, shape = 56, rate = 21)
x_future <- rpois(U, lambda = lambda_post)
mean(x_future >= 4)
```

Estimate: ≈ 0.28 (exact value from the negative binomial predictive is 0.2802). There is about a 28% posterior predictive probability that the company receives at least 4 complaints on a future day.